IGCSE 0607 Practice Questions — E1.2 — Combined Questions & Markscheme

Sets

Total Questions: 6 — **Total Marks:** 36 Marks

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- **E1.2** Sets
  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(4 Qs)*

1 $\mathsf{U} = \{x : x \text{ is an integer and } 1 \leq x \leq 10\}$
$A = \{x : x \text{ is a prime number}\}$
$B = \{x : x \text{ is a factor of } 12\}$

(a) List the members of set A.

..... **[1]**

(b) List the members of set B.

..... **[1]**

(c) Find $n(A \cup B)$.

..... **[1]**

(d) Write down the elements of $A \cap B$.

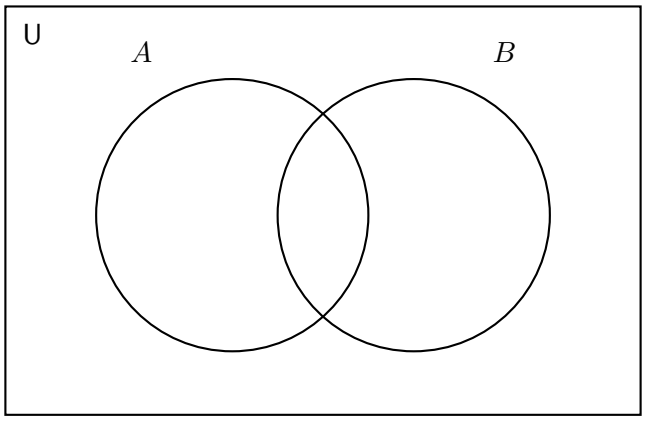
..... **[1]**

(e) Write down a number that is in U but is not in $A \cup B$.

..... **[1]**

2 Let U be the set of integers from 1 to 10 inclusive.
Let A be the set of prime numbers.
Let B be the set of odd numbers.

(a) Complete the Venn diagram to show the elements of these sets.



[2]

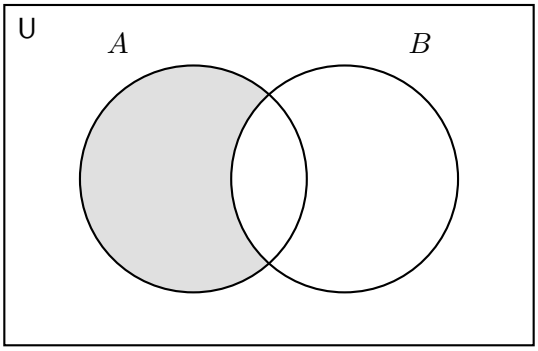
(b) List the elements of $A \cap B'$.

..... **[1]**

(c) Find $n(A \cup B)$.

..... **[1]**

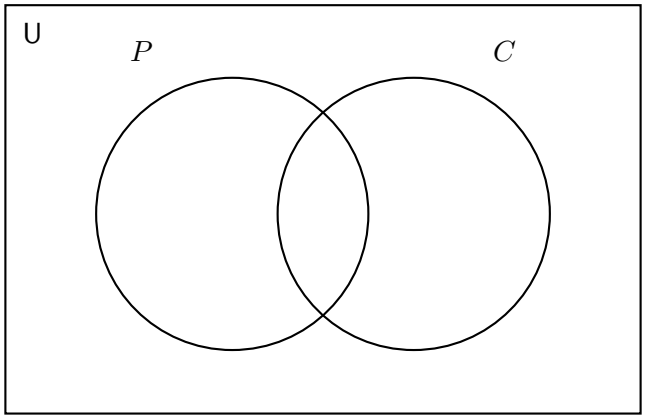
(d) Write down the set notation for the shaded region in the Venn diagram above.



..... **[1]**

3 There are 45 students in a class.
Let U be the set of students in this class.
Let P be the set of students who study Physics.
Let C be the set of students who study Chemistry.
$n(P) = 28$
$n(C) = 22$
$n(P \cap C) = x$
$n((P \cup C)') = 4$

(a) Complete the Venn diagram in terms of x.



[2]

(b) Form an equation in x and solve it to find the value of x.

$x =$ ..... **[2]**

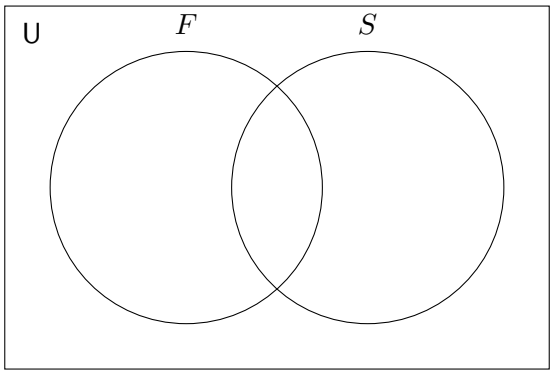
(c) Find $n(P \cap C')$.

..... **[1]**

4 In a class of 40 students:

- 25 study French (F)
- 18 study Spanish (S)
- 5 study neither French nor Spanish

(a) Complete the Venn diagram to show this information.



[3]

(b) Find $n(F \cap S')$.

..... **[1]**

(c) A student is chosen at random from the class.
Find the probability that they study exactly one of the two languages.

..... **[2]**

(d) Two students are chosen at random from those who study French.
Find the probability that both also study Spanish.

..... **[1]**

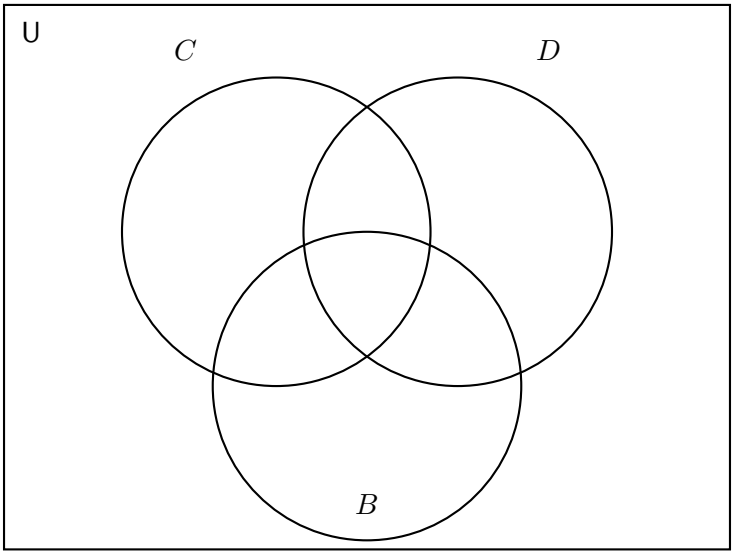
5 In a survey of 100 people, information was collected about the pets they own.

Let U be the set of people surveyed.
Let C be the set of people who own a cat.
Let D be the set of people who own a dog.
Let B be the set of people who own a bird.

- 8 people own all three pets.
- 12 people own a cat and a dog but not a bird.
- 7 people own a cat and a bird but not a dog.
- 10 people own a dog and a bird but not a cat.
- 3 people do not own any of these pets.

The number of people who own only a cat is x.
The number of people who own only a bird is also x.
The number of people who own only a dog is $2x$.

(a) Complete the Venn diagram to show this information.



[3]

(b) Find the value of x.

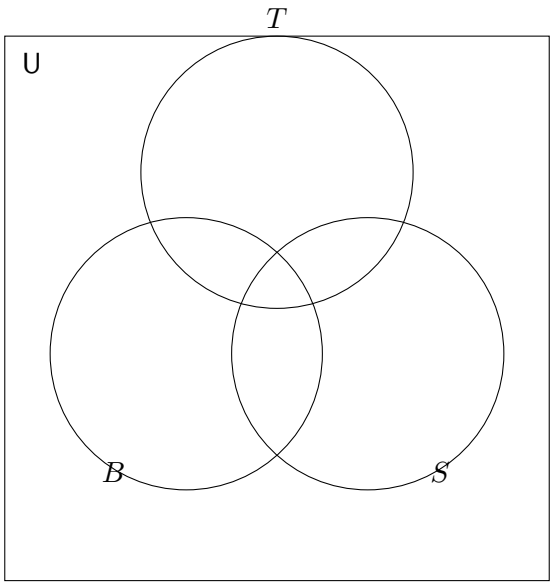
$x =$ ..... **[2]**

(c) Find $n((C \cup D) \cap B')$.

..... **[2]**

6 There are 60 people in a sports club.

- 30 play Tennis (T)
- 28 play Badminton (B)
- 25 play Squash (S)
- 12 play both Tennis and Badminton
- 9 play both Tennis and Squash
- 11 play both Badminton and Squash
- 4 play all three sports



(a) Complete the Venn diagram.

[3]

(b) Find $n(T \cup B \cup S)'$.

..... **[1]**

(c) Find $n(T \cap (B \cup S))$.

..... **[1]**

(d) Use set notation to describe the region containing the number 9 (the number of people who play Badminton only).

..... **[2]**

1 **(a)** 2, 3, 5, 7 **[B1]**

(b) 1, 2, 3, 4, 6, 12 **[B1]**

(c) 8 **[B1]**

(d) 2, 3 **[B1]**

(e) 8 or 9 or 10 **[B1]**

[Back to Question] *[Total Marks: 5]*

2 **(a)** Intersection: $3, 5, 7$ **[B1]**

Remainder of diagram correct: A only region has 2; B only region has $1, 9$; Outer region has $4, 6, 8, 10$ **[B1]**

(b) 2 **[B1]**

(c) 6 **[B1]**

(d) $A \cap B'$ **[B1]**

[Back to Question] *[Total Marks: 5]*

3 **(a)** Intersection: x and Outer region: 4 **[B1]**

P only region: $28 - x$ AND C only region: $22 - x$ **[B1]**

(b) $(28 - x) + x + (22 - x) + 4 = 45$ oe **[M1]**

$x = 9$ **[A1]**

(c) 19 **[B1]**

[Back to Question] *[Total Marks: 5]*

4 **(a)** 5 in outer region **[B1]**

8 in intersection **[B1]**

17 in F only AND 10 in S only **[B1]**

(b) 17 **[B1]**

(c) $\frac{27}{40}$ **[B2]**

M1 for $17 + 10$ or 27 seen **[M1]**

(d) $\frac{8}{25} \times \frac{7}{24}$ **[M1]**

$\frac{7}{75}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

5 **(a)** Intersection of all three: 8 AND Outer region: 3 **[B1]**

Two-set only intersections: 12, 7, and 10 in correct regions **[B1]**

Single set regions: x (in C), $2x$ (in D), and x (in B) **[B1]**

(b) $x + 2x + x + 8 + 12 + 7 + 10 + 3 = 100$ oe **[M1]**

$x = 15$ **[A1]**

(c) Identifies the regions required: Only C + Only D + ($C \cap D$ only) OR uses $x + 2x + 12$ **[M1]**

57 **[A1]**

[Back to Question] *[Total Marks: 7]*

6 **(a)** 4 in center intersection **[B1]**

8, 5, 7 in the two-sport intersections **[B1]**

13, 9, 9 in the one-sport regions AND 5 in outer region **[B1]**

(b) 5 **[B1]**

(c) 17 **[B1]**

(d) $B \cap T' \cap S'$ or $B \cap (T \cup S)'$ **[B2]**

M1 for $B \cap \ldots$ **[M1]**

[Back to Question] *[Total Marks: 7]*